

Does an Industry Effect Exist for Initial Public Offerings?

Aigbe Akhigbe

University of Akron

Stephen F. Borde*

University of Central Florida

Ann Marie Whyte

University of Central Florida

Abstract

We examine the impact of initial public offerings (IPOs) on rival firms and find that the valuation effects are insignificant. This insignificant reaction can be explained by offsetting information and competitive effects. Significant positive information effects are associated with IPOs in regulated industries and the first IPO in an industry following a period of dormancy. Significant negative competitive effects are associated with larger IPOs in competitive industries, those in relatively risky industries, those in high-performing industries, and those in the technology sector. IPO firms that use the proceeds for debt repayment appear to represent a more significant competitive threat to rival firms relative to IPO firms that use their proceeds for other purposes.

Keywords: Initial public offering, IPO, intra-industry effects, competitive effects, information effects

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*Corresponding author: Department of Finance, College of Business Administration, University of Central Florida, Orlando, FL 32816-1400; Phone: (407) 823-2977; Fax: (407) 823-6676; E-mail: stephen.borde@bus.ucf.edu

1. Introduction

The literature presents clear evidence that initial public offerings (IPOs) of equity are underpriced. For example, Ibbotson and Ritter (1995) observe that IPO underpricing for the period from 1960 to 1992 is about 15%, on average. However, existing studies focus primarily on the valuation effects for the issuing firm. We examine whether the announcement of an IPO also has industry-wide implications. We are motivated to consider the issue by previous studies which have shown that firm-level announcements have implications for rival firms. For example, Lang and Stulz (1992) and Cheng and McDonald (1996) find that bankruptcy announcements have significant implications for rival firms. Similarly, Slovin, Sushka, and Poloncheck (1992) document significant intra-industry effects for seasoned equity issues by commercial and investment banks, Akhigbe, Madura, and Whyte (1997) for bond rating downgrades resulting in significant negative effects for rivals, and Kohers (1999) for dividend omissions. Other studies document insignificant intra-industry effects on average. For example, Laux, Starks, and Yoon (1998) find that large dividend changes have diverse effects within the same industry, resulting in insignificant average price changes, and Hertzl (1991) finds insignificant announcement period abnormal returns for rivals in response to stock repurchases. Similarly, Howe and Shen (1998) find that dividend initiations do not contain an industry-wide component. Instead, the information conveyed by dividend initiations is firm-specific, resulting in no discernible intra-industry effect.

In this context, we test the hypothesis that an IPO may have significant intra-industry implications. An IPO is likely to have industry-wide implications for at least two reasons. First, an IPO could signal a change in the outlook for the industry as a whole, resulting in significant valuation effects for rival firms.¹ Second, the IPO could cause a reassessment of the competitive balance within the industry. Since the IPO raises funds that the firm can use to expand and compete more effectively, the firm could gain a competitive advantage over its rivals. Indeed, Slovin, Sushka, and Ferraro (1995) find that rivals react negatively to the 107 IPOs in their sample. They conclude that the negative reaction occurs because the IPO conveys unfavorable information to the market regarding industry prospects. However, their primary focus is on rival reaction to equity carve-outs, spinoffs, and asset sell-offs. Further, they do not explore underlying differences in the characteristics of the IPO or the industry that could cause the reaction of rivals to vary. We build on their work using a more comprehensive sample to explicitly examine the factors that explain the diverse valuation effects of rival firms.

We find that the valuation effects for rival firms are insignificant, seeming to indicate that IPOs are purely firm-specific events with no implications for rivals.

¹ Alternatively, one might argue that a favorable industry outlook triggers an attractive environment for a firm in that industry to go public.

However, we find that the insignificant reaction can be explained by the presence of offsetting positive information and negative competitive effects. Significant positive information effects are associated with IPOs in regulated industries and the first IPO in the industry following a period of dormancy. Significant negative competitive effects are associated with relatively large IPOs in highly competitive industries, those in relatively risky industries, those in high-performing industries, and those in the technology sector. IPO firms that use the proceeds for debt repayment appear to represent a more significant competitive threat to rival firms relative to firms that use their proceeds for other purposes.

2. Hypotheses

The impact of an IPO on rival firms may vary, depending on several factors. First, an IPO could convey information that is relevant to the industry as a whole, causing a revaluation of rival firms' share prices (information effects). Second, an IPO could alter the competitive balance within the industry, causing the market to reassess the relative competitiveness of rival firms (competitive effects). Finally, an IPO may have an insignificant impact on a sample of rivals, either because the event is firm specific or has a diverse impact on rivals, resulting in insignificant effects on average.

2.1. Information effects

Ritter (1991) argues that firms may choose to go public when investors are “irrationally over-optimistic” about the future of certain industries. Thus, an IPO could be construed as a favorable signal for the industry as a whole, causing rival firms' share prices to be revised upwards. Conversely, Slovin, Sushka, and Ferraro (1995) note that managerial signaling models developed by Leland and Pyle (1977) and Titman and Trueman (1986), among others, are built on the assumption that there is an adverse selection problem associated with the decision to go public. Given that managers of the IPO firm are presumed to be better informed than outside investors, outside investors infer negative signals from the decision to go public. If the information is pertinent to the industry as a whole, the share prices of rival firms should fall. The following factors are examined to test for information effects.

Relative size of rival firms: Following Atiase (1985), we hypothesize that the strength of the signal conveyed by an IPO is inversely related to the relative size of rivals. Since smaller firms are not as closely followed by analysts, there is generally less information available about smaller firms. Thus, the incremental information conveyed by the IPO is likely to cause a more significant revaluation in the share prices of relatively small rivals.

Regulation in the industry: Kohers (1999) argues that industries that are relatively homogeneous may have a more pronounced reaction to intra-industry information transfer. One measure of the degree of homogeneity is the presence of regulation.

Kohers notes that the constraints imposed by regulation result in less diversity across firms, increasing the propensity of investors to draw industry-wide inferences from firm-level events such as IPOs. Thus, we expect IPOs by firms in regulated industries to have stronger information effects for rivals.

Stock exchange of IPO: The exchange on which shares of the IPO firm trade may be an important determinant of rival valuation effects. One school of thought suggests that NYSE-traded firms have a higher level of institutional ownership and are generally more actively traded, causing investors to assign greater importance to information related to such firms (Kohers, 1999). Another school of thought suggests that the opposite may be true. This is based on the assertion that there is a greater degree of information asymmetry regarding NASDAQ-traded firms, which could cause the incremental information released by the IPO to be more valuable. Thus, the relation between the stock exchange of the IPO and rival valuation effects is an empirical question.

First IPO in the industry: Previous intra-industry studies (Akhigbe, Madura, and Whyte, 1997) have shown that the first event is generally associated with the strongest rival valuation effects. Thus, we expect that the first IPO in an industry following a period of dormancy should have the strongest information effects. The advent of a new IPO generates additional exposure, both for the IPO firm and the industry as a whole. To the extent that an IPO conveys information that is relevant to the industry as a whole, the first IPO should result in more pronounced valuation effects for rivals relative to subsequent IPOs.

2.2. Competitive effects

The decision to go public could cause investors to reevaluate the competitive position of the IPO firm relative to its rivals. Since the IPO increases the visibility of the firm and raises funds that the firm can use to expand and compete more effectively, the firm could gain a competitive advantage. These changes are likely to affect rivals adversely. The following factors are examined to test for competitive effects.

Growth opportunities within the industry: The extent of growth opportunities within the industry could affect the ability of a rival firm to respond to the competitive threat of a newly public firm. Firms with relatively high Tobin's q ratios are likely to have more investment opportunities than those with relatively low q ratios (Lang, Stulz, and Walkling, 1991). In the context of the entire industry, rivals with relatively low q ratios should have fewer growth prospects and may not have the ability to respond to the competitive threat of a newly public firm with greater resources. Consequently, they are more likely to be adversely affected by an IPO.

Degree of concentration within the industry: We expect that rivals in less concentrated (more competitive) industries will experience more severe competitive effects (Lang and Stulz, 1992). The additional capital raised by the IPO firm is more likely to be a threat to rivals in industries with a relatively low degree of concentration.

Interaction of degree of concentration and IPO size: We expect that relatively large IPOs in industries with a low degree of concentration (highly competitive industries) should have the strongest competitive effects. The greater the size of the IPO, the more funds available to the firm and the greater the potential competitive threat. This threat would be exacerbated in an industry that is already highly competitive.

Use of proceeds: The proceeds generated by an IPO may be used for different purposes including debt repayment, expansion, and acquisition financing. On one hand, use of IPO proceeds for purposes such as financing an expansion could generate a stronger competitive advantage for the IPO firm. Alternatively, use of the proceeds for debt repayment could enhance the firm's competitive position to the extent that its debt burden is reduced. In this case, rivals may react more adversely to the IPO.

Relative risk level of rival firms: Rivals with a relatively high level of risk could be limited in their ability to challenge their competitors or respond appropriately to the competitive threat. Thus, to the extent that the risk level of rivals is higher than that of the IPO firm, the competitive effects should be stronger.

Prior performance of rival firms: Underperforming rivals are more likely to be adversely affected by a new IPO, because they may not have the capacity to respond appropriately to the additional competition. Alternatively, high-performing rivals could lose because firms tend to enter industries with excess rents. As more competition enters the industry, the market anticipates that excess rents will be reduced, causing high-performing rivals to react negatively.

Technology IPO: Kohers and Kohers (2000) find that high-tech targets are perceived as relatively attractive investments because of the higher growth potential of technology firms. In general, the high technology industry is characterized by a high degree of competition, a rapidly changing environment, and a high level of uncertainty. In this environment, the entrance of a newly public firm with additional resources would place rivals at a competitive disadvantage, causing them to react more adversely to an IPO.

Degree of similarity: Lang and Stulz (1992), Firth (1996), and Erwin and Miller (1998) argue that the degree of similarity between a firm and its rivals may influence the intra-industry effects. Thus, we hypothesize that rivals whose returns are relatively highly correlated with the IPO firm are more likely to be impacted by the IPO.

2.3. Insignificant effects

Two interpretations of an insignificant impact on rivals are plausible. First, the event is firm-specific with no corresponding implications for rivals. Evidence of this type of firm-specific event is presented by Hertzal (1991) for stock repurchases and Howe and Shen (1998) for dividend initiations. Second, the event could have heterogeneous effects within the industry. In this case, the IPO could have industry-wide implications that are being obscured by offsetting information and competitive

Table 1

Distribution of IPOs (1989–2000)

This table presents the number of IPOs per year and the percentage of the total number of IPOs occurring each year. The sample of IPOs is obtained from Thomson Financial.

Year	All IPO filings	% of total	IPO filings with CRSP SIC code and at least one publicly traded rival firm	
				% of total
1989	15	0.38	10	0.40
1990	119	3.05	65	2.61
1991	291	7.45	182	7.30
1992	354	9.06	206	8.26
1993	509	13.03	292	11.71
1994	341	8.73	219	8.78
1995	399	10.22	293	11.75
1996	586	15.00	408	16.37
1997	413	10.57	258	10.35
1998	242	6.20	135	5.42
1999	432	11.06	319	12.80
2000	205	5.25	106	4.25
TOTAL	3,906	100.00	2,493	100.00

effects. For example, Laux, Starks, and Yoon (1998) find that the information and competitive effects offset, resulting in insignificant returns on average to rivals of firms announcing large dividend revisions.

3. Sample selection and methodology

The sample of all IPO filings occurring between 1989 and 2000 is obtained from Thomson Financial. The sample consists of 3,906 IPO filings occurring on the NYSE, AMEX, and NASDAQ that have data available from the Center for Research in Security Prices (CRSP).² When we impose the requirement that each IPO firm have at least one publicly traded rival with the same Standard Industry Classification (SIC) code, the sample is reduced to 2,493 IPOs. Table 1 shows the distribution of all IPO filings and IPO filings with at least one publicly traded rival. The number of IPO filings ranges from a minimum of 15 in 1989 to a maximum of 586 in 1996. The sample of IPO filings with at least one publicly traded rival ranges from a low of ten in 1989 to a high of 408 in 1996.

² We exclude spinoffs from the sample because some capital generally flows to the parent company, which could confound the results.

3.1. Assessing rival abnormal returns

We obtain abnormal returns (ARs) for rival firms using the standard event study methodology. The ARs are assessed on both the filing date and the initial trading date. Since the filing date is the date on which public information first becomes available about the new competition, rivals are likely to react on the filing date. Rivals could also react when the IPO first begins trading, since additional information may be released during that period. Following Lang and Stulz (1992), we form an equally-weighted portfolio of publicly traded rival firms that share the same Standard Industrial Classification (SIC) code as the IPO firm on both the filing date and the initial trading date (day 0). This procedure accounts for potential cross-sectional correlation of returns in the industry. We use the returns of each rival portfolio to estimate the market model parameters over the period (–220 to –20) relative to the IPO filing (or trading) date and compute the ARs of each rival portfolio (p) and for each day in the event period (–11, +11). We report cumulative abnormal returns (CARs) over three different time intervals: day (0), days (0, +1), and days (+2, +10). We then compute the equally-weighted average of the CARs of the rival portfolios. Following the methodology of Mikkelsen and Partch (1988), we use the z-statistic to test the statistical significance of the CARs.

3.2. Cross-sectional model development

We use the following cross-sectional model, which has the advantage of controlling for both competitive and information effects simultaneously (see, for example, Howe and Wei, 1993) to explain the reaction of rival firms:

$$\begin{aligned} RIVCAR_j = & \beta_0 + \beta_1 RELSIZE_j + \beta_2 REGUL_j + \beta_3 EXCH_j \\ & + \beta_4 FIRST_j + \beta_5 GROWTH_j + \beta_6 HI_j + \beta_7 HISIZE_j \\ & + \beta_8 USE_j + \beta_9 RELRISK_j + \beta_{10} PRIORPERF_j \\ & + \beta_{11} TECH_j + \beta_{12} CORR_j + \beta_{13} TIME_j + \varepsilon_j \end{aligned} \quad (1)$$

where:

$RIVCAR_j$ = the two-day (0, +1) CAR of the portfolio of rival firms in response to an IPO filing by firm j

$RELSIZE_j$ = the relative size of rival firms measured as the median market value of rivals relative to the market value of the IPO firm. Market values are calculated as the product of the number of common shares outstanding and the closing price on the first trading day

$REGUL_j$ = 1 if the IPO firm is in a regulated industry, including financial services, utility, communications, and transportation; 0 otherwise

- $EXCH_j$ = 1 if the IPO firm is traded on the NYSE or AMEX; 0 otherwise
- $FIRST_j$ = 1 if the IPO is the first in the industry within a two-year period of time prior to the current IPO; 0 otherwise
- $GROWTH_j$ = 1 if the median value of the ratio of market to book value of common stock of all rival firms, a proxy for Tobin's q , is greater than 1; 0 otherwise.
- HI_j = the Herfindahl index for the industry which is computed as the squared sum of the fractions of industry market capitalization of rival firms
- $HISIZE_j$ = an interaction term calculated by multiplying the Herfindahl Index for the industry by the relative size of the IPO
- USE_j = 1 if the proceeds are used for debt repayment; 0 otherwise
- $RELRISK_j$ = the relative beta of rival firms compared to the beta of IPO firm j (measured from +20 to +220). This variable assumes a value of 1 when the industry median beta is greater than the beta for the IPO firm; 0 otherwise
- $PRIORPERF_j$ = the CAR of rival firms in the 60-day period prior to the IPO filing by firm j
- $TECH_j$ = 1 if the IPO firm j is in the technology industry;³ 0 otherwise
- $CORR_j$ = 1 if the correlation of the returns between the IPO firm and the industry portfolio (during the event period +20 to +220) is greater than the median; 0 otherwise
- $TIME_j$ = 1 if the IPO of firm j occurs in 1991 or earlier; 0 otherwise
- ε_j = a random error term

The TIME variable is included to capture any changes in rival reaction in the post-1991 period, the period not covered by the Slovin, Sushka, and Ferraro (1995) study. Beyond this control function, the TIME variable reflects two years with relatively low IPO volume, as shown in Appendix A. Following the explanation posed with respect to the FIRST variable, we argue that in a cold IPO market and in an industry with little IPO activity, a filing is more newsworthy, causing more information to be disseminated. This could lead to positive information effects for rivals.

We estimate three versions of the above model. The first model includes the factors related to information effects, the second includes the factors related to competitive effects, and the third model includes both sets of factors.

³ Garner, Nam, and Ottoo (2002) point out that SIC codes sometimes incorrectly classify high-tech firms. To overcome this potential bias, we identify technology firms using the Value Line industry classifications. Value Line continually updates the industry groups to reflect new entrants, allowing us to include firms from relatively new industries such as internet and biotechnology. Other technology industries include telecommunications, computer hardware and software, aerospace, data processing, and office equipment. See Kohers and Kohers (2000) for a comprehensive list of technology industries.

Table 2

Comparison of characteristics of IPO firms and their industry rivals

This table provides mean and median values for various characteristics of the IPO firms and the industry rivals. The market value of equity is measured 220 trading days following the IPO; the proxy for Tobin's Q is the ratio of market to book value of common stock; ROE is the return on equity. Free cashflow is measured as (earnings before interest and taxes plus depreciation)/total assets. Leverage is the ratio of total debt to total assets. Beta is estimated over the period starting 20 days after the IPO and ending 220 days after the IPO. The mean and median market values of equity and beta are calculated from the CRSP database (N = 2493). The other variables are obtained from the Compustat database (N = 2433). Financial statement data are obtained in the calendar year end following the IPO. The *t*-statistic tests whether the mean and median values are significantly different for the IPO firm and the industry portfolios.

Variable	IPO firms		Industry portfolios		<i>t</i> -test for difference in	
	Mean	Median	Mean	Median	Mean	Median
1. Market value of equity	307.24	103.43	365.20	56.73	−0.96	1.30
2. Total assets	711.76	50.36	393.40	51.74	1.15	−0.01
3. Tobin's Q	4.16	3.05	2.72	2.35	15.97***	6.76***
4. ROE (%)	8.05	9.14	10.27	11.99	−5.37***	−4.52***
5. Free cashflow (%)	2.98	8.57	7.97	8.71	−10.69***	−0.49
6. Leverage (%)	16.65	4.26	13.42	5.39	0.96	−0.91

*** Indicates statistical significance at the 0.01 level.

4. Empirical results

4.1. Valuation effects for rivals

Table 2 provides a comparison of the characteristics of IPO firms relative to their rivals.⁴ Rivals and IPO firms are similar in terms of market value of equity, total assets, and leverage. However, IPO firms have a significantly higher Tobin's Q (4.16 on average compared to 2.72 for rival firms) and beta (1.63 on average compared to 1.33 for rivals), but IPO firms have a significantly lower return on equity (ROE) and free cash flow.

Although not reported in the table, the mean (median) number of rivals per event is 11.91 (5) respectively with a minimum of one rival and a maximum of 224 rivals per event. The sample consists of firms from 190 different industries. Appendix A reports the 190 different four-digit SIC codes and the number of IPOs each year.

Table 3 discloses CARs for rivals. Panel A reports the results for rival firms on the IPO filing date. Rival firms do not experience a significant reaction to IPO filings,

⁴ Although not reported in the tables, we also estimate the degree of underpricing for the IPO firm and find that the average underpricing is about 23%. This is consistent with previous studies that have documented significant underpricing. We estimate the underpricing as follows: $UP_j = (P_1 - P_0)/P_0$ where UP_j measures the extent of the underpricing, P_1 is the closing price on the first trading day, and P_0 is the offer price. We then compute the equally-weighted average underpricing for the portfolio of IPO firms on the initial trading date of the IPO. Closing share prices are obtained from the CRSP daily returns file.

Table 3

Cumulative abnormal returns (CARs) for rivals (N = 2493)

This table presents the cumulative abnormal returns (CARs) of industry rivals in response to the IPO. Rival portfolios contain all rival firms (identified using the CRSP SIC code) that are publicly traded at the time of the IPO. The sample period is from 1989 to 2000. Rival CARs are calculated as the differences between the actual returns and expected returns. Expected returns are generated from the market model parameters, estimated with daily returns from the period $t-220$ to $t-20$ where t_0 is date of the IPO. The z -statistic tests the null hypothesis that the CARs equal zero.

Panel A: CARs for rival portfolios on the IPO filing date

Event period	Number of events	CAR (%)	z -statistic	% positive
(0)	2,493	0.04	0.27	47
(0, +1)	2,493	-0.07 ^a	-0.38	45
(+2, +10)	2,493	-0.08	-0.97	45

Panel B: CARs for rival portfolios on the initial trading date of the IPO

Event period	Number of events	CAR (%)	z -statistic	% positive
(0)	2,493	-0.05	-1.63	46
(0, +1)	2,493	-0.02 ^b	-0.73	45
(+2, +10)	2,493	0.10	0.27	47

^aThe median, maximum, and minimum day (0) (day (0, +1)) abnormal returns for rival firms are 0.00% (-0.20%), 84.00% (89.81%), and -30.28% (-39.03%) respectively.

^bThe median, maximum, and minimum day (0) (day (0, +1)) abnormal returns for rival firms are -0.18% (-0.29%), 49.81% (41.63%), and -15.28% (-33.84%) respectively.

regardless of the time interval considered. Similar results are shown in Panel B for rival portfolios on the initial trading date.⁵ These results differ from those of Slovin, Sushka, and Ferraro (1995), who find a significant negative reaction for rivals. The difference may be related to differences in sample size, sample composition, and time period studied. Specifically, their sample examines an earlier period (from 1980 to 1991), consists of relatively large firms, uses a considerably smaller sample ($n = 107$), and excludes financial firms and firms traded on NASDAQ, where most IPOs are launched.

Table 4 shows the CARs for rival portfolios based on factors related to information and competitive effects. The results for some variables are presented by quartiles, while others are presented in two different subgroups. In general, the results are not significantly different across subgroups for most of the factors considered, with two notable exceptions: the first IPO in an industry (FIRST) and prior performance

⁵The lack of a significant reaction on both the filing and initial trading dates may be related to the fact that although the filing conveys information to the market, additional information is released during the subsequent road show. Further, a substantial amount of time may elapse between the filing and initial trading dates, making it difficult to capture a significant reaction within the event window.

Table 4

CARs for subsamples of industry portfolios partitioned by characteristics

This table shows CARs for subgroups based on various characteristics of the IPO, IPO firm, rivals, and the industry as a whole. The *z*-statistic tests the significance of the CARs and the *t*-statistic tests whether the CARs are significantly different from each other (and from first to fourth quartiles). RELSIZE is the size of rival firms relative to the market value of the IPO firm; REGUL is equal to 1 if the IPO firm is in a regulated industry, 0 otherwise; EXCH is equal to 1 if the IPO firm is traded on the NYSE or AMEX, 0 otherwise; FIRST is equal to 1 if the IPO is the first in the industry within a two-year period of time prior to the current IPO; 0 otherwise; GROWTH is equal to 1 if the median value of market to book value of common stock of all rival firms of firm *j*, a proxy for Tobin's *q*, is greater than 1, 0 otherwise; HI is the Herfindahl index for the industry; USE is equal to 1 if the proceeds are used for debt repayment, 0 otherwise; RELRISK is the median beta of rival firms relative to the beta of firm *j*; PRIORPERF is the CAR of rival firms in the 60-day period prior to the IPO filing by firm *j*; TECH is equal to 1 if the IPO firm *j* is in the technology industry, 0 otherwise; and CORR is equal to 1 if the correlation between the returns of the IPO firm and the industry portfolio is greater than the median; 0 otherwise. TIME is equal to 1 if the IPO filing date of firm *j* occurs in 1991 or earlier, 0 otherwise. For those variables broken down by quartile, the smallest or lowest values are represented in the first quartile, and the largest or highest values are represented in the fourth quartile.

	Quartile or group	N	CAR	<i>z</i> -statistic	<i>t</i> -statistic
RELSIZE	1	608	−0.01	−0.02	
	2	608	−0.03	−0.07	
	3	608	−0.07	0.07	
	4	609	−0.19	−0.83	0.58
REGUL	1	211	0.22	0.58	
	0	2,222	−0.10	−0.60	0.98
EXCH	1	247	0.30	1.49	
	0	2,186	−0.12	−0.92	1.52
FIRST	1	250	0.45	1.18	
	0	2,183	−0.14	−0.87	1.86*
GROWTH	1	1,873	−0.05	0.20	
	0	560	−0.15	−1.28	0.34
HI	1	608	−0.04	−0.10	
	2	608	0.06	−0.52	
	3	608	0.05	−0.19	
	4	609	−0.39	−0.01	1.21
USE	1	448	−0.21	−0.62	
	0	1,985	−0.04	−0.16	0.67
RELRISK	1	608	0.01	0.49	
	2	608	−0.08	−0.39	
	3	608	0.01	−0.33	
	4	609	−0.20	−0.52	0.88
PRIORPERF	1	608	0.54	2.81***	
	2	608	−0.06	−0.14	
	3	608	−0.22	−2.04**	
	4	609	−0.57	−1.98**	3.29***
TECH	1	817	−0.13	0.05	
	0	1,616	−0.04	−0.55	0.47
CORR	1	1,197	−0.01	−0.81	
	0	1,236	−0.13	0.22	0.63
TIME	1	237	0.33	0.16	
	0	2,196	−0.12	−0.49	0.98

*** Indicates statistical significance at the 0.01 level.

** Indicates statistical significance at the 0.05 level.

* Indicates statistical significance at the 0.10 level.

of rival firms (PRIORPERF). The first IPO in an industry generates higher rival CARs (0.45%) relative to subsequent IPOs (−0.14%), although the difference is only marginally significant at the 0.10 level. With respect to PRIORPERF, the results vary significantly across subgroups. Rivals with relatively low prior performance (quartile 1) gain 0.54% in response to the IPO filing, significant at the 0.01 level. Rivals in the higher-performing quartiles (quartiles 3 and 4) experience significantly negative CARs. The *t*-statistic (which tests whether the returns for quartiles 1 and 4 are significantly different from each other) reveals that the difference is significant at the 0.01 level. This finding is contrary to our hypothesis that underperforming rivals are more likely to be adversely affected, since they may not have the resources to respond to the competitive threat. The inconsistency in results across rivals with differing prior performance may be explained as follows. Since firms tend to enter industries with excess rents, more competition within those industries could signal excess rent reductions. This could have a more adverse effect on high-performing rivals, because their high performance may be partially driven by excess rents. Thus, one could argue that high-performing rivals have more to lose from the entrance of a strong competitor, and low-performing rivals may benefit more from additional information released by the IPO. Furthermore, high-performing rivals may already have been privy to the information revealed by the IPO, resulting in poor-performing rivals gaining more from new information than they lose from competitive effects. Thus, prior performance may proxy for both a competitive and an information effect, depending on the prior condition of the rival firm.

Table 5 shows the CARs for three cross-classifications of rival portfolios: CORR and HI, Tobin's Q and HI, and IPO size and HI. The CARs are insignificant for most subgroups with the exception of rivals with a low Tobin's Q and a low HI. The significant negative CAR of 0.30% suggests that rivals in low growth industries that are not highly concentrated (i.e. more competitive industries) could be at a disadvantage when a competitor goes public. However, the *t*-statistic indicates no significant difference between the CARs of rivals at the two extremes: low Tobin's Q and low HI compared to high Tobin's Q and high HI.

4.2. Robustness tests

The analysis of CARs reported thus far is based on using an equally-weighted portfolio of rival returns. Following Erwin and Miller (1998), we test the robustness of our results by reestimating the CARs using a value-weighted portfolio of rival returns. The results are qualitatively similar to those obtained using the equally-weighted portfolio and, hence, are not reported. The similarity of the results using value- and equally-weighted portfolios is consistent with Lang and Stulz (1992) and Erwin and Miller (1998). As an additional robustness test, we use the methodology employed by Szewczyk (1992) to limit the number of rivals. We form industry portfolios by randomly selecting a maximum of five rivals (the median number of rivals in the

Table 5

Two-way classification of rival portfolio CARS

This table shows the results for rival portfolios based on various cross-classifications. HI is the Herfindahl Index for the industry, CORR is the correlation between the returns of the IPO firm and the industry portfolio, the proxy for Tobin's Q is the ratio of market to book value of common stock, and IPO size is the size of the IPO.

Panel A: Based on correlation and HI

	Low HI			High HI			<i>t</i> -statistic
	N	CAR (%)	<i>z</i> -statistic	N	CAR (%)	<i>z</i> -statistic	
Negative CORR	624	−0.01	0.26	612	−0.26	0.06	
Positive CORR	627	0.02	−0.84	570	−0.05	−0.27	0.18

Panel B: Based on Tobin's Q and HI

	Low HI			High HI			<i>t</i> -statistic
	N	CAR (%)	<i>z</i> -statistic	N	CAR (%)	<i>z</i> -statistic	
Low Tobin's Q	197	−0.30	−2.07**	363	−0.06	−0.14	
High Tobin's Q	1,054	0.07	0.33	819	−0.21	−0.07	0.29

Panel C: Based on IPO size and HI

	Low HI			High HI			<i>t</i> -statistic
	N	CAR (%)	<i>z</i> -statistic	N	CAR (%)	<i>z</i> -statistic	
Relatively small IPOs	843	0.02	−0.66	600	−0.33	−0.85	
Relatively large IPOs	408	−0.02	0.16	582	0.01	0.73	0.05

** Indicates statistical significance at the 0.05 level.

unrestricted sample) with the same four-digit SIC code as the IPO firm. The results indicate that rival CARs are not sensitive to the selection process used to determine the rivals.

4.3. Cross-sectional results

Table 6 shows the cross-sectional results based on the CARs derived from IPO filing dates. Model 1 shows the results when only the factors related to information effects are included in the model. The primary result that emerges from this table is that the information effects are generally positive and statistically significant, with the exception of the relative size of rival firms (RELSIZE) and the exchange of the IPO firm (EXCH). The positive coefficient estimate on REGUL is consistent with Kohers's (1999) argument that industries characterized by a high degree of homogeneity should have a more pronounced reaction to intra-industry information transfer.

Table 6

Explaining cross-sectional variation in rival valuation effects

This table shows the results of estimating three models: the information effects model, the competitive effects model, and the combined model. The dependent variable is the 2-day (0, +1) CAR of the portfolio of rival firms in response to an IPO filing by firm *j*. RELSIZE is the size of rival firms relative to the market value of the IPO firm; REGUL is equal to 1 if the IPO firm is in a regulated industry, 0 otherwise; EXCH is equal to 1 if the IPO firm is traded on the NYSE or AMEX, 0 otherwise; FIRST is equal to 1 if the IPO is the first in the industry within a two-year period of time prior to the current IPO; 0 otherwise; GROWTH is equal to 1 if the median value of market to book value of common stock of all rival firms of firm *j*, a proxy for Tobin's *q*, is greater than 1, 0 otherwise; HI is the Herfindahl index for the industry; HISIZE is an interaction term calculated by multiplying the HI for the industry by the relative size of the IPO; USE is equal to 1 if the proceeds are used for debt repayment, 0 otherwise; RELRISK is the median beta of rival firms relative to the beta of firm *j*; PRIORPERF is the CAR of rival firms in the 60-day period prior to the IPO filing by firm *j*; TECH is equal to 1 if the IPO firm *j* is in the technology industry, 0 otherwise; and CORR is equal to 1 if the correlation between the returns of the IPO firm and the industry portfolio is greater than the median; 0 otherwise. TIME is equal to 1 if the IPO filing date of firm *j* occurs in 1991 or earlier, 0 otherwise. Variance inflation factors, which test for multicollinearity, are generally low (<1.5), indicating that multicollinearity is not a problem. Marquardt and Snee (1975) suggest that a VIF value of greater than 5.0 indicates the presence of multicollinearity.

Variable	Model 1: Information effects	Model 2: Competitive effects	Model 3: Competitive and information effects
Intercept	−0.016***	0.018***	0.013**
RELSIZE	0.001		−0.001
REGUL	0.020***		0.035***
EXCH	0.008		0.012*
FIRST	0.013**		0.022***
GROWTH		−0.000	0.000
HI		−0.013***	−0.018***
HISIZE		−0.003***	−0.003***
USE		−0.029***	−0.028***
RELRISK		−0.010***	−0.010**
PRIORPERF		−0.009***	−0.015***
TECH		−0.019***	−0.014***
CORR		0.013***	0.016***
TIME	0.029***	0.031***	0.020**
	R ² = 0.017	R ² = 0.067	R ² = 0.085
	Adjusted R ² = 0.015	Adjusted R ² = 0.063	Adjusted R ² = 0.080
	F-Value = 8.32***	F-Value = 19.14***	F-Value = 17.19***
	N = 2433	N = 2433	N = 2433

*** Indicates statistical significance at the 0.01 level.

** Indicates statistical significance at the 0.05 level.

* Indicates statistical significance at the 0.10 level.

The implication is that the constraints imposed by regulation increase the propensity of investors to draw industry-wide inferences from firm-level events such as IPOs. The first IPO (FIRST) within an industry also generates the strongest information effects.

Model 2 presents the results for factors related to competitive effects. It is again worth noting that the coefficient estimates on the competitive factors are generally negative and statistically significant, with the exception of GROWTH and CORR. The degree of concentration (HI) is inversely related to rival valuation effects, consistent with our hypothesis that rivals in less concentrated (more competitive) industries experience more severe competitive effects. The results suggest that the additional capital raised by the IPO firm is more likely to be a threat to rivals in industries with a relatively low degree of concentration. When we interact the size of the IPO with the degree of concentration in the industry (HISIZE), the reaction is also significant. The negative coefficient estimate suggests that large IPOs in highly competitive industries have the strongest competitive effects. A large IPO generates more funds for the newly public firm, increasing the threat to rivals, and this threat appears to be exacerbated in industries that are already highly competitive. The coefficient estimate of -0.029 for the USE variable is negative and significant at the 0.01 level. This suggests that when IPO proceeds are used for debt repayment the firm could be more of a competitive threat to rivals, perhaps because management may be preparing for a more aggressive stance in the wake of a reduced debt burden. The coefficient estimate on RELRISK (-0.010) is negative and significant at the 0.01 level, suggesting that competitive effects are more pronounced for higher risk rivals that may not have the flexibility to respond to the competitive threat. The prior performance of rivals (PRIORPERF) is inversely related to the reaction of rival firms. This is contrary to the expected reaction and suggests that underperforming firms may have less to lose from the entrance of a newly public competitor. The competitive effects are significant for IPOs in the technology sector. The coefficient estimate on TECH is -0.019 , and is significant at the 0.01 level, supporting the notion that technology rivals are more vulnerable to competitive shocks, perhaps because of the rapidly changing, highly competitive environment in which these firms operate. Finally, the CARs are positively related to CORR and TIME.

Model 3 shows the results when both information and competitive factors are included in the same model. The results are qualitatively similar to those reported when the models are estimated separately.

We also estimate a simple regression model (not reported in the tables) with the degree of underpricing of the IPO firm as the independent and rival CARs from the initial trading date (as opposed to the filing date) as the dependent variable. Since the underpricing is only apparent on the trading date, the CARs on the filing date (which occurs well before the underpricing is evident) would not be appropriate as the dependent variable on that date. We estimate the model in an effort to test the hypothesis that IPOs that are significantly underpriced provide more information regarding industry prospects. The coefficient estimate on the degree of underpricing is positive (0.0063) and significant (at the 0.05 level: $t = 2.25$), suggesting that more information is released by highly underpriced IPOs.

5. Conclusion

We examine a comprehensive sample of IPOs to determine whether IPOs have implications for rival firms. We find that the average valuation effects for rival firms are insignificant. The insignificant reaction does not indicate that IPOs are purely firm-specific events with no implications for the industry as a whole. Instead, we find that the insignificant reaction can be explained by the presence of offsetting positive information and negative competitive effects. Significant positive information effects are associated with IPOs in regulated industries and the first IPO in the industry following a period of dormancy. Significant negative competitive effects are associated with relatively large IPOs in highly competitive industries, those in relatively risky industries, those in better-performing industries, and those in the technology sector. Firms that use the proceeds of the IPO for debt repayment appear to represent a more significant competitive threat to rival firms relative to firms that use their proceeds for other purposes.

Appendix A

Number of IPOs by year and industry (4-digit SIC code)

This appendix shows the 190 different SIC codes represented in the sample during the sample period 1989–2000.

Number	SIC Code	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
1	1321	0	0	0	0	0	0	0	1	0	0	0	0
2	1380	0	1	1	0	1	0	1	3	2	0	0	0
3	1531	0	0	0	1	0	0	0	0	1	0	0	0
4	1620	0	0	0	1	0	0	0	0	0	0	0	0
5	1790	0	0	0	0	0	1	0	0	0	0	0	0
6	2020	1	0	0	1	0	0	0	0	1	1	0	0
7	2030	0	0	0	0	1	2	0	0	1	0	0	0
8	2052	0	0	0	0	0	1	0	0	0	0	0	0
9	2060	0	1	0	0	2	0	0	1	0	0	0	0
10	2080	0	0	0	0	1	0	0	0	0	0	0	0
11	2250	0	0	0	1	0	0	0	0	0	0	0	0
12	2260	0	0	0	1	0	0	0	0	0	0	0	0
13	2360	0	0	0	0	1	0	0	0	0	0	0	0
14	2430	0	0	0	0	1	0	0	0	0	0	0	0
15	2520	0	0	1	0	0	0	0	0	0	0	0	0
16	2650	0	0	0	1	0	0	0	1	0	0	0	0
17	2711	0	0	0	0	1	0	0	0	1	0	0	0
18	2730	0	0	1	1	0	0	0	1	0	0	0	0
19	2750	0	1	0	1	0	1	0	0	2	1	1	0
20	2760	0	0	0	0	0	0	1	0	0	0	0	0
21	2819	0	1	0	0	0	1	0	0	0	0	0	0
22	2830	2	2	34	15	12	11	12	36	11	2	8	2
23	2834	1	0	0	1	0	0	0	1	5	0	0	14
24	2835	0	0	0	0	1	0	0	1	0	0	0	1

(continued)

Appendix A (continued)

Number of IPOs by year and industry (4-digit SIC code)

Number	SIC Code	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
25	2840	0	0	0	0	0	1	1	0	0	0	0	0
26	2870	0	0	0	0	1	1	0	0	0	0	0	0
27	2879	0	0	1	0	0	0	0	0	0	0	0	1
28	2890	0	0	0	1	0	1	1	0	0	0	0	0
29	2911	0	1	0	0	0	0	0	0	0	0	0	0
30	3070	0	0	0	0	1	0	0	0	0	0	0	0
31	3080	0	0	2	1	4	1	0	1	2	1	0	0
32	3144	0	0	0	1	1	0	0	0	0	0	0	0
33	3149	0	1	0	0	0	0	0	0	0	0	0	0
34	3310	0	0	0	0	2	1	1	1	0	0	0	0
35	3312	0	0	0	0	1	1	0	0	1	0	0	0
36	3320	0	0	0	0	0	1	0	0	0	0	0	0
37	3356	0	0	0	0	0	0	0	1	0	0	0	0
38	3429	0	0	0	0	0	1	0	0	0	0	0	0
39	3440	0	0	0	0	1	1	1	1	2	0	0	0
40	3448	0	0	0	1	0	0	0	0	0	0	0	0
41	3460	0	0	0	0	2	0	0	0	0	0	0	0
42	3470	0	0	0	0	0	0	1	0	0	0	0	0
43	3510	0	0	0	0	1	0	0	0	0	0	0	0
44	3519	0	0	0	1	0	0	0	0	0	0	0	0
45	3530	0	0	0	0	0	1	1	1	0	1	0	0
46	3550	0	0	0	2	4	2	0	0	0	0	0	0
47	3560	1	0	2	1	1	1	1	1	1	0	0	0
48	3570	0	1	4	10	9	5	15	11	3	0	13	5
49	3571	0	0	0	0	0	0	1	0	0	0	1	1
50	3572	0	0	0	0	1	0	0	1	0	0	1	2
51	3577	0	0	1	2	0	1	0	1	0	0	0	2
52	3579	0	0	0	0	0	0	1	0	0	0	0	0
53	3580	0	0	0	1	4	1	1	0	0	0	0	0
54	3610	0	0	0	0	1	0	0	0	0	0	0	1
55	3620	0	0	0	1	0	1	1	2	0	0	1	1
56	3630	0	0	0	0	1	0	0	0	0	0	0	0
57	3650	0	0	1	2	0	2	2	0	0	0	0	0
58	3651	0	0	0	0	0	0	0	0	0	0	1	0
59	3660	0	2	4	3	10	10	14	15	5	4	16	0
60	3661	0	0	0	0	3	0	0	0	0	1	1	9
61	3663	0	0	0	0	0	4	4	1	0	1	1	4
62	3670	0	2	7	8	19	11	19	6	16	5	8	2
63	3674	0	3	0	0	1	1	2	0	1	0	3	8
64	3679	0	0	1	0	0	0	0	0	0	0	0	0
65	3690	0	0	1	3	3	3	2	3	0	0	0	0
66	3710	0	0	1	1	3	6	3	4	0	0	0	0
67	3711	0	0	1	0	1	0	0	0	0	0	0	0
68	3714	0	0	0	0	0	0	0	0	2	0	0	0
69	3720	0	0	0	0	0	0	0	2	1	1	0	0

(continued)

Appendix A (continued)

Number of IPOs by year and industry (4-digit SIC code)

Number	SIC Code	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
70	3730	0	0	0	0	0	0	1	0	0	1	0	0
71	3810	0	1	1	0	1	0	1	1	1	0	0	0
72	3812	0	0	0	0	0	1	0	0	0	0	0	0
73	3820	0	0	2	7	7	2	5	7	10	0	4	1
74	3829	0	0	0	0	0	0	0	1	0	0	0	0
75	3840	0	4	17	17	6	8	22	31	12	0	1	0
76	3842	0	0	1	2	0	0	1	0	0	0	0	2
77	3845	0	0	0	0	0	0	0	0	0	0	0	3
78	3851	0	0	0	0	0	1	1	0	0	0	0	0
79	3860	0	0	0	0	0	0	0	1	0	0	0	0
80	4010	0	0	1	0	1	0	0	2	0	0	0	0
81	4011	0	0	0	1	0	0	0	0	0	0	0	0
82	4210	0	1	1	1	1	9	2	0	4	0	0	0
83	4213	0	0	0	0	1	0	0	0	0	0	0	0
84	4510	0	1	0	0	2	2	1	0	1	1	0	0
85	4730	0	0	0	1	1	0	1	1	0	1	0	0
86	4810	0	1	1	3	4	4	11	9	5	4	23	0
87	4812	0	0	0	0	0	0	0	1	0	0	0	5
88	4813	0	0	0	0	0	0	0	1	1	1	2	3
89	4830	0	0	1	0	4	1	4	0	0	0	0	0
90	4832	0	0	1	0	0	0	0	1	0	0	0	0
91	4840	0	0	0	0	4	1	2	1	1	0	6	0
92	4911	0	0	1	0	0	0	0	1	0	0	0	2
93	4920	0	0	0	0	0	0	0	0	1	0	0	0
94	4930	0	0	0	0	0	0	0	1	0	0	0	0
95	4950	0	3	2	2	1	0	0	2	2	2	0	0
96	4953	0	2	0	0	0	0	0	0	1	0	0	0
97	4960	0	0	1	0	0	0	0	0	0	0	0	0
98	5020	0	1	0	0	0	0	0	0	0	0	0	0
99	5030	0	0	0	0	1	0	0	1	0	0	0	0
100	5040	0	1	1	4	3	4	2	7	2	0	1	0
101	5045	0	0	0	0	0	0	1	0	0	0	0	0
102	5060	1	3	1	0	3	7	1	2	0	0	1	0
103	5090	0	0	0	0	6	0	4	2	4	0	0	0
104	5110	0	0	1	0	0	1	0	1	0	0	0	0
105	5120	0	0	1	1	1	1	1	1	1	0	0	0
106	5130	0	0	1	1	1	2	2	3	1	1	0	0
107	5140	0	0	1	1	2	0	0	2	1	0	0	0
108	5160	0	0	0	2	0	0	0	0	1	0	0	0
109	5190	0	0	2	0	2	0	0	2	2	1	0	0
110	5311	0	0	1	2	0	0	0	0	0	0	1	0
111	5411	0	0	1	0	0	0	1	1	0	0	0	0
112	5620	0	0	1	0	3	0	0	0	0	1	0	0
113	5650	0	0	1	0	1	0	1	1	1	0	0	0
114	5660	0	0	0	0	1	1	0	0	1	0	0	0

(continued)

Appendix A (continued)

Number of IPOs by year and industry (4-digit SIC code)

Number	SIC Code	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
115	5710	0	1	0	0	1	1	1	0	0	0	0	0
116	5712	0	0	0	0	1	0	0	0	0	0	0	0
117	5730	0	0	1	1	5	1	1	2	2	3	2	0
118	5800	0	0	0	0	0	1	0	0	0	0	0	0
119	5810	0	2	2	9	8	6	7	10	5	3	2	0
120	5812	0	0	2	0	1	0	0	1	1	0	0	0
121	5910	0	0	0	1	0	0	1	0	0	0	1	0
122	5940	0	0	0	3	2	1	1	3	1	1	6	0
123	5944	0	0	0	0	0	0	0	1	0	0	0	0
124	5960	1	2	1	1	2	1	3	4	2	2	7	0
125	5990	0	0	3	6	5	4	1	3	0	0	0	0
126	6020	0	0	1	2	6	3	2	5	4	16	3	0
127	6021	0	0	0	0	0	0	0	1	0	0	0	0
128	6022	0	0	0	1	0	0	0	0	0	0	0	1
129	6030	1	1	2	1	2	0	0	4	3	3	1	0
130	6159	0	1	1	1	1	0	0	2	0	2	0	0
131	6162	0	0	1	0	0	0	0	0	0	0	0	0
132	6210	0	0	1	1	1	0	0	2	0	1	0	0
133	6211	0	0	1	1	1	0	0	2	1	1	2	0
134	6282	1	0	0	4	2	0	0	0	1	1	1	0
135	6310	0	0	0	1	0	0	1	0	0	0	0	0
136	6311	0	0	1	1	0	0	1	1	0	0	0	0
137	6320	0	2	2	1	3	1	2	0	1	0	0	0
138	6321	0	0	1	0	0	0	0	0	0	1	1	0
139	6330	0	1	0	2	3	0	1	1	1	1	0	0
140	6331	0	1	4	1	0	0	2	1	0	0	0	0
141	6410	0	0	1	0	0	2	0	2	0	1	0	0
142	6411	0	0	0	0	0	0	0	1	0	0	1	0
143	6510	0	0	0	1	0	0	0	0	0	0	0	0
144	6550	0	0	0	1	0	0	0	1	1	0	0	0
145	6710	1	2	0	2	2	3	1	2	1	0	1	0
146	6712	0	0	0	0	0	0	0	0	0	0	0	2
147	6720	0	0	0	1	1	1	1	1	1	0	0	0
148	6790	0	0	2	1	0	0	0	2	5	0	0	0
149	6798	0	0	1	5	24	15	2	7	19	2	0	0
150	7010	0	0	1	0	3	4	0	4	0	0	0	0
151	7011	0	0	1	0	1	1	3	4	0	1	0	0
152	7200	0	0	1	1	0	1	1	1	0	0	0	0
153	7299	0	0	1	0	1	0	0	0	0	0	0	0
154	7310	0	1	2	0	0	0	3	4	0	0	0	0
155	7312	0	0	0	0	0	0	0	1	0	0	0	0
156	7330	0	0	0	0	1	0	1	1	0	0	0	0
157	7350	0	0	0	2	0	1	1	0	1	1	0	0
158	7359	0	0	0	1	2	0	1	1	1	3	0	0
159	7360	0	0	0	1	3	1	1	4	1	1	0	0

(continued)

Appendix A (continued)

Number of IPOs by year and industry (4-digit SIC code)

Number	SIC Code	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
160	7363	0	0	0	0	0	1	1	0	0	0	0	0
161	7370	0	2	15	19	28	18	65	85	43	40	120	8
162	7372	0	2	4	4	6	3	7	11	8	6	29	18
163	7373	0	0	1	2	1	1	1	5	5	2	6	5
164	7374	0	1	1	2	1	1	1	0	2	1	0	0
165	7380	0	1	1	0	2	5	5	11	8	4	27	0
166	7389	0	0	0	0	0	0	1	1	3	0	0	0
167	7390	0	0	0	0	0	0	0	2	0	0	0	0
168	7810	0	1	2	0	1	3	3	4	1	0	0	0
169	7812	0	0	0	0	0	0	0	1	1	0	0	0
170	7820	0	0	0	0	1	0	0	0	1	0	0	0
171	7822	0	0	0	0	0	0	0	0	0	0	1	0
172	7832	0	0	0	0	0	0	0	0	1	0	0	0
173	7940	0	0	0	0	0	1	0	0	0	0	0	0
174	7990	0	0	0	4	6	5	1	3	0	0	3	0
175	8011	0	1	1	0	0	0	2	0	0	0	0	0
176	8020	0	0	0	0	0	0	0	1	0	0	0	0
177	8070	0	2	3	3	1	1	1	2	0	0	0	0
178	8071	0	0	1	0	1	1	0	0	0	0	0	1
179	8080	0	0	2	2	0	3	1	3	0	0	0	0
180	8090	0	4	5	4	3	4	5	6	5	1	1	0
181	8093	0	1	0	0	0	0	0	0	0	0	1	0
182	8099	0	1	1	1	0	0	0	0	0	0	0	0
183	8240	0	0	0	0	0	1	0	0	0	0	0	0
184	8710	0	0	1	0	1	0	0	1	0	0	1	0
185	8711	0	0	1	0	1	0	0	0	0	0	0	0
186	8730	0	0	3	2	2	3	8	7	8	3	2	0
187	8740	0	1	1	0	1	0	1	4	7	3	4	0
188	8742	0	0	0	0	1	0	0	0	1	0	1	2
189	8744	0	0	0	0	0	0	0	1	0	0	0	0
190	8990	0	0	0	1	0	0	0	0	1	0	1	0
Total		10	65	182	206	292	219	293	408	258	135	319	106

References

- Akhigbe, A., J. Madura, and A.M. Whyte, 1997. Intra-industry effects of bond rating adjustments, *Journal of Financial Research* 20, 545–561.
- Atiase, R., 1985. Predisclosure information, firm capitalization, and security price behavior around earnings announcements, *Journal of Accounting Research* 23, 21–36.
- Cheng, L.T.W. and J.E. McDonald, 1996. Industry structure and ripple effects of bankruptcy announcements, *The Financial Review* 31, 783–807.
- Erwin, G.R. and J.M. Miller, 1998. The intra-industry effects of open market share repurchases: Contagion or competitive? *Journal of Financial Research* 21, 389–406.
- Firth, M., 1996. Dividend changes, abnormal returns, and intra-industry firm valuations, *Journal of Financial and Quantitative Analysis* 31, 189–210.

- Garner, J.L, J. Nam, and R.E. Ottoo, 2002. Determinants of corporate growth opportunities of emerging firms, *Journal of Economics and Business* 54, 73–93.
- Hertzel, M.G., 1991. The effects of stock repurchases on rival firms, *Journal of Finance* 46, 707–716.
- Howe, J.S. and Y. Shen, 1998. Information associated with dividend initiations: Firm-specific or industry-wide? *Financial Management* 27, 17–26.
- Howe, J.S. and P. Wei, 1993. The valuation effects of warrant extensions, *Journal of Finance* 48, 305–314.
- Ibbotson, R. and J. Ritter, 1995. Initial public offerings, in *North-Holland Handbooks of Operations Research and Management Science–Finance*, ed. Robert Jarrow, Vojislav Maksimovic, and William Ziemba (North-Holland, New York).
- Kohers, N., 1999. The Industry-wide implications of dividend omission and initiation announcements and the determinants of information transfer, *The Financial Review* 34, 137–158.
- Kohers, N. and T. Kohers, 2000. The value creation potential of high-tech mergers, *Financial Analysts Journal* 56, 40–50.
- Lang, L. and R.M. Stulz, 1992. Contagion and competitive intra-industry effects of bankruptcy announcements, *Journal of Financial Economics* 32, 45–60.
- Lang, L.H.P., R.M. Stulz, and R.A. Walkling, 1991. A test of the free cash flow hypothesis: The case of bidder returns, *Journal of Financial Economics* 29, 315–335.
- Laux, P., L.T. Starks, and P. Yoon, 1998. The relative importance of competition and contagion in intra-industry information transfers, *Financial Management* 27, 5–16.
- Leland, H.E. and D. Pyle, 1977. Informational asymmetries, financial structure, and financial intermediation, *Journal of Finance* 32, 371–387.
- Mikkelsen, W. and M. Partch, 1988. Withdrawn security offerings, *Journal of Financial and Quantitative Analysis* 23, 119–34.
- Ritter, J., 1991. The long-run performance of initial public offerings, *Journal of Finance* 46, 3–27.
- Slovin, M.B., M.E. Sushka, and S.R. Ferraro, 1995. A comparison of the information conveyed by equity carve-outs, spin-offs, and asset sell-offs, *Journal of Financial Economics* 37, 89–104.
- Slovin, M.B., M.E. Sushka, and J.A. Polonchek, 1992. Informational externalities of seasoned equity issues, *Journal of Financial Economics* 32, 87–101.
- Szewczyk, S., 1992. The intra-industry transfer of information inferred from announcements of corporate security offerings, *Journal of Finance* 47, 1935–1945.
- Thomson Financial (data services).
- Titman, S. and B. Trueman, 1986. Information quality and the valuation of new issues, *Journal of Accounting and Economics* 8, 159–172.

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